

Total No. of Questions : 8]

SEAT No. :

PA-2641

[Total No. of Pages : 2

[5927]-421

B.E. (Mechanical Engineering)

PRODUCT DESIGN AND DEVELOPMENT

(2019 Pattern) (Semester - VII) (Elective - IV) (402045A)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates :

- 1) Solve Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicate full marks.
- 4) Use of electronic pocket calculator is allowed.
- 5) Assume suitable data, if necessary.

- Q1) a)** List down different methods used for product teardown process and explain any one. [7]
- b) Describe in detail benchmarking. [6]
- c) What is concept analysis? List down different aspects of concept analysis. [4]

OR

- Q2) a)** What is concept selection? Explain Pugh's chart with example. [7]
- b) Write a short note on FAST method. [6]
- c) What is product policy of an organization? List down various product policies. [4]

- Q3) a)** What is product modularity? Explain types of Modularity. [7]
- b) Explain BOM with example. [6]
- c) What is Tolerance? Describe the types of tolerances. [4]

OR

- Q4) a)** What is product architecture? Explain types of product architecture. [7]
- b) What is dimensioning? Describe arrangement of dimensioning. [6]
- c) What is Fit? Describe the types of Fits. [4]

P.T.O.

- Q5) a)** List down different methods of economic analysis of product and explain break even analysis. [8]
- b)** What is Rapid prototyping? Define and enlist various methods of prototyping. [6]
- c)** Define letter of intent, purchase order and product costing in vendor development. [4]

OR

- Q6) a)** Explain stereo lithography in detail with suitable sketch. [8]
- b)** What is production capacity planning? Explain the steps followed in planning. [6]
- c)** Why homologation certificate is important in design and development? Explain with example. [4]

- Q7) a)** Write a short note on APQP. [8]
- b)** Write a short note on DFMEA. [6]
- c)** Discuss the elements of PLM in detail. [4]

OR

- Q8) a)** List down types of FMEA and explain any one with example. [8]
- b)** Write a short note on PDM. [6]
- c)** Discuss design for robustness in detail. [4]

